

Demonstration of Proposed Features

Date	15 March 2026
Team ID	TM-OPTICROP-04
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Soil & Telemetry Dashboard	Real-time visualization of soil moisture, NPK levels, and temperature parameters ingested from IoT sensor nodes.	Implemented	Yes	Successfully integrated with dynamic web dashboard.
2	Crop Yield Predictive Engine	ML optimization model (XGBoost) forecasting yields based on soil composition, weather metrics, and historic data.	Implemented	Yes	Achieved validation accuracy of 92.4% during tests.
3	Optimal Irrigation Scheduler	Recommendation module suggesting irrigation runtimes based on dynamic evapotranspiration rates.	Implemented	Yes	Provides hourly automated water usage strategies.
4	Dynamic Weather Alerts	Broadcaster module ingesting dynamic open weather API forecasts to emit instant crop protection warnings.	Implemented	Yes	Fully synchronized with SMS threshold notifications.
5	Soil Fertilizer Optimizer	AI prescription system matching soil deficits with exact, cost-efficient fertilizer composition balances.	Implemented	Yes	Successfully generated 15 test recommendations.

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6	Disease/Pest Identification	Computer vision classifier (ResNet50 backend) detecting leaf health anomalies from user photo uploads.	Partial	Yes	Inference engine working; cloud scaling optimization pending.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	5
Total Features Demonstrated	6
Overall Implementation Rate (%)	83.33%